REPORT

Forest Fire Detection & Spread Prediction System

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Milestone – 1

Problem Statement

Forest fires are increasing due to rising temperatures and changing climate conditions. Traditional fire detection methods rely on satellite monitoring or manual inspection, which can result in delayed detection. This delay allows small fires to spread rapidly, causing severe environmental damage, loss of biodiversity, and economic loss. Therefore, a real-time monitoring system is required to detect fire early and predict its spread effectively.

Overview of System Design

The first milestone focuses on designing an IoT-based forest fire detection and prediction system. The system uses sensors to continuously monitor environmental conditions such as temperature, humidity, gas levels, and flame presence.

An ESP32 microcontroller collects and processes sensor data, which is then transmitted to the cloud using WiFi. The cloud platform stores and visualizes this data, enabling real-time monitoring and analysis. This design provides a scalable and efficient solution for early fire detection and prediction.

System Architecture Description

The system follows a layered architecture consisting of four main layers:

1. Sensing Layer

This layer includes sensors such as:

- DHT11 (Temperature & Humidity)
- MQ-2 (Gas/Smoke detection)
- Flame Sensor (Fire detection)

These sensors collect environmental data continuously.

2. Processing Layer

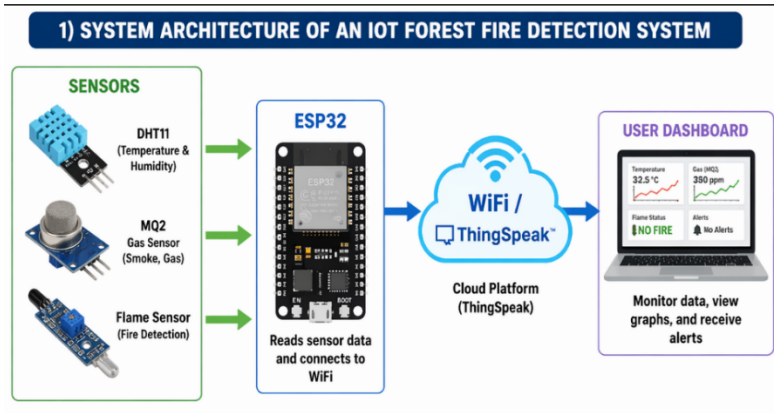
The ESP32 microcontroller reads sensor data, processes it, and applies fire detection logic. It also calculates a risk score for fire spread prediction based on environmental conditions.

3. Cloud/Storage Layer

The processed data is sent to the ThingSpeak cloud platform. The data is stored with timestamps, enabling continuous logging and future analysis.

4. Visualization Layer

The data is displayed using graphs and dashboards on ThingSpeak. This allows users to monitor fire conditions and trends remotely in real time.



Simulation Setup and Implementation

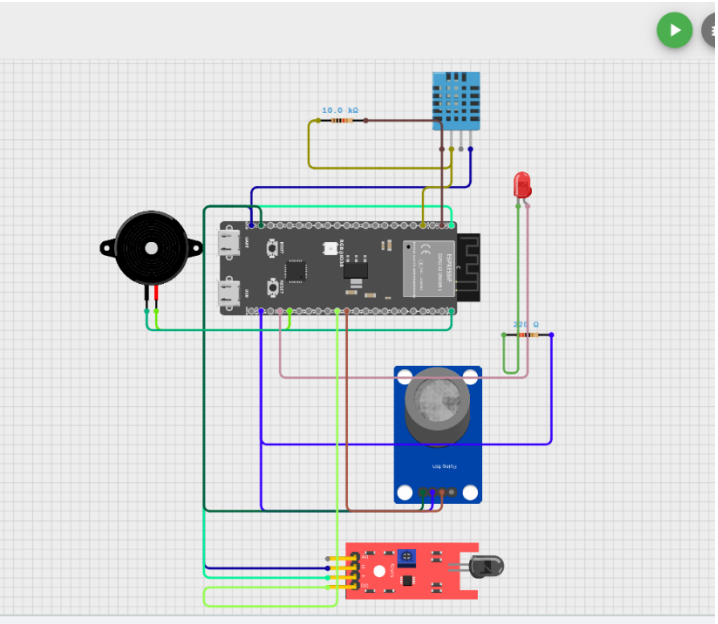
The system was simulated using Cirkit Designer to validate connections and functionality before hardware implementation.

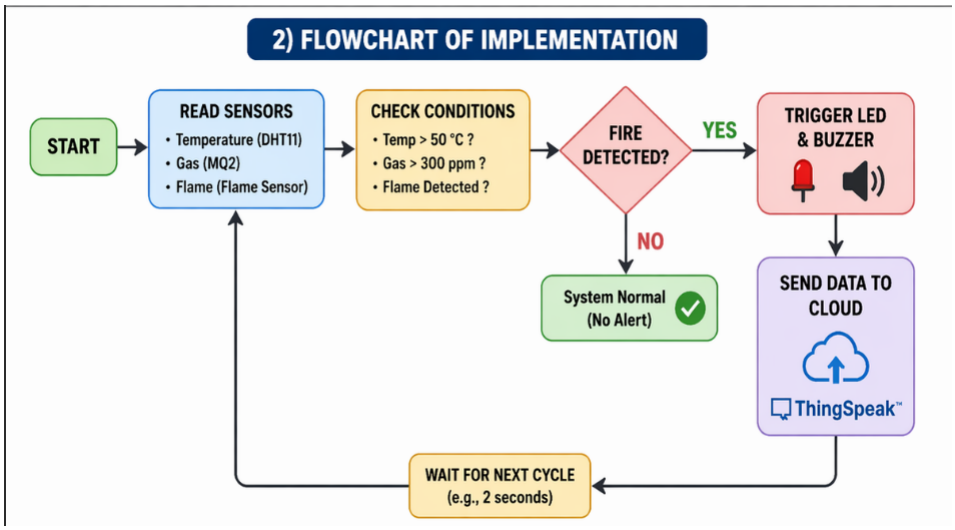
All components including ESP32, DHT11, MQ-2, and flame sensor were connected virtually. Proper pin configurations, power supply connections, and communication methods were ensured.

The simulation allowed real-time monitoring of sensor values and helped verify:

- Correct wiring
- Sensor behavior
- Data acquisition

Fire detection logic was implemented, where alerts are triggered when abnormal conditions are detected. This step ensured the system design functional and reliable.





Cloud Integration using ThingSpeak

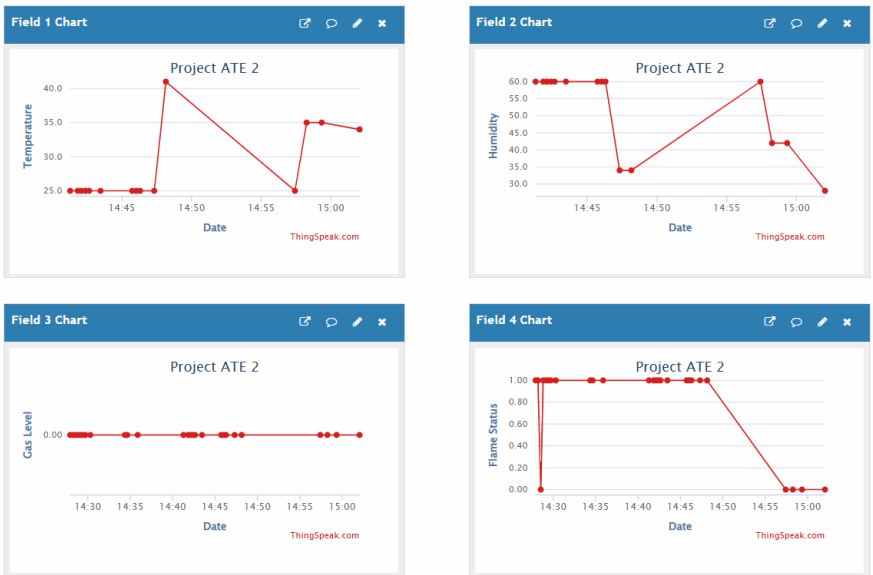
To enable real-time monitoring, the system integrates with the ThingSpeak cloud platform.

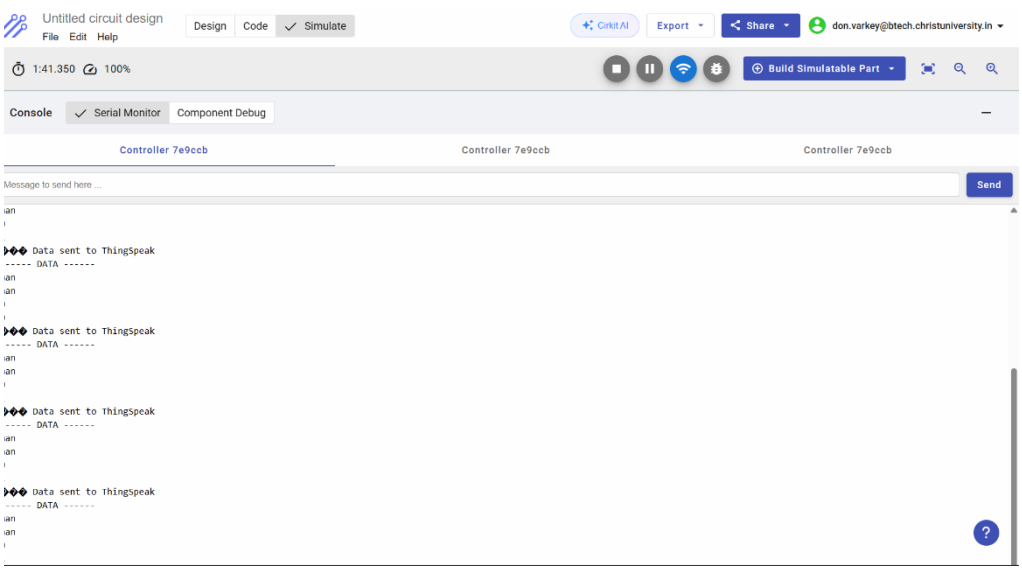
The ESP32 sends sensor data such as temperature, humidity, gas levels, and flame status to ThingSpeak using WiFi. Each parameter is assigned to specific fields in the channel.

ThingSpeak stores the data and provides graphical visualization, allowing users to observe trends and variations. The platform also supports data export for further analysis and machine learning.

This integration enables:

- Remote monitoring
- Real-time data visualization
- Continuous data storage





Outcome

At the end of this milestone, a complete system architecture has been successfully designed. The simulation setup has been implemented and validated, and real-time data transmission to ThingSpeak has been achieved.

The system is capable of detecting fire conditions and provides a foundation for fire spread prediction. This milestone establishes the base for further development, including machine learning models and advanced analytics in future stages.